

Descriptive:

1. Compiler: Definition, different phases, compiler vs interpreter
2. Lexical Analyzer: working process, different components, tokens
3. Syntax Analyzer: Grammars and trees, ambiguity, transition diagrams
4. CFG: definition, different components
5. Intermediate Code: Semantic tree, Quadruples vs triples vs indirect triples.

Mathematical:

1. Generate regular expressions for matching certain string formats
2. Draw transition diagrams for how RegEx matches a string
3. Check a grammar for ambiguity
4. Eliminate left recursion and left factoring from a given grammar
5. Remove all types of ambiguity from a grammar (remove op ambiguity, remove left recursion, implement left factoring)
6. Generate three address code from statements and represent using quadruples/triples/indirect triples